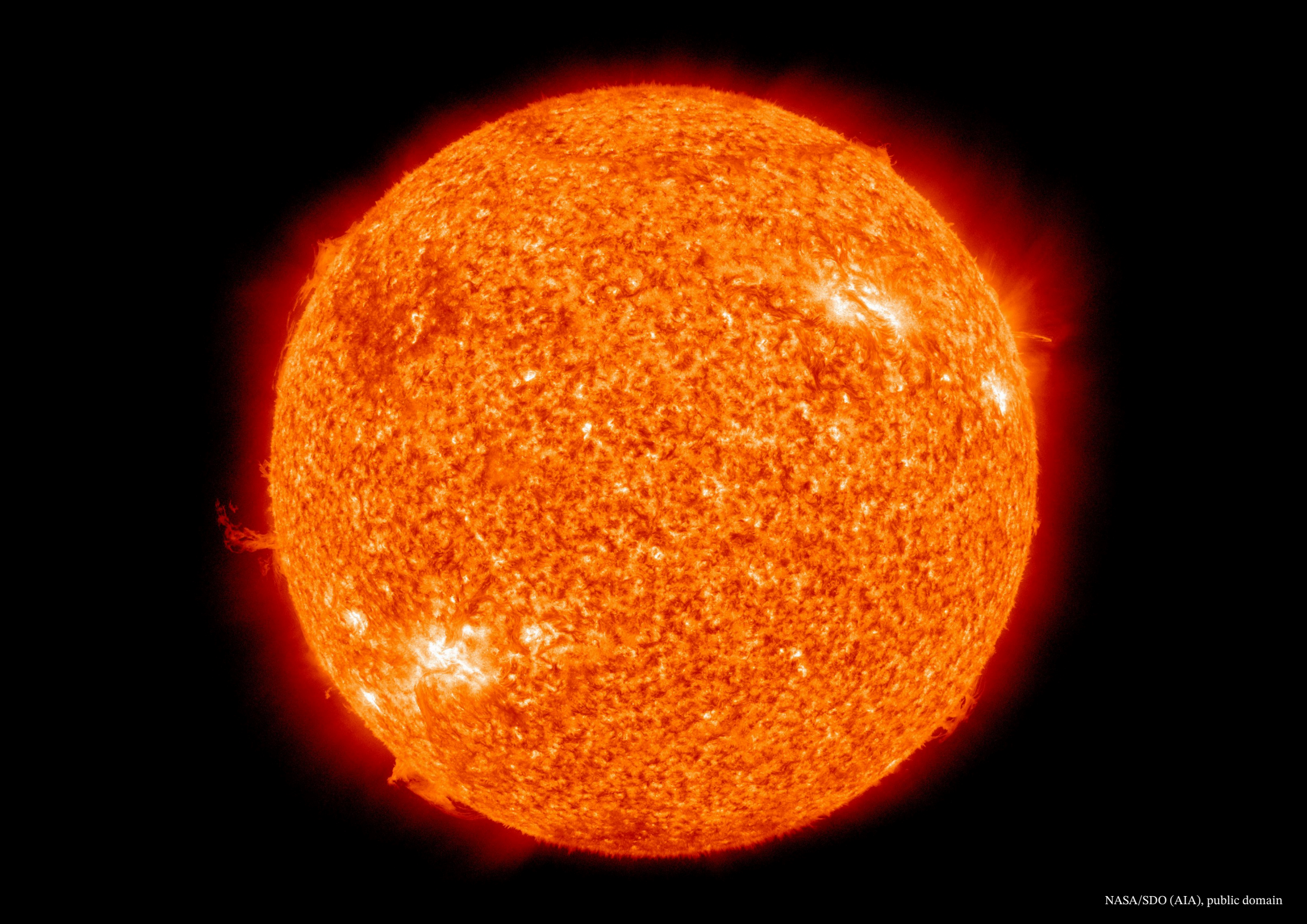
Fusion in stars

Christopher Albert



NASA/SDO (AIA), public domain

3.1 Energy budget of the Sun

Sun $L_\odot = 3.83 \times 10^{26}\ \mathrm{W}, \quad M_\odot = 1.99 \times 10^{30}\ \mathrm{kg}, \quad R_\odot = 6.96 \times 10^{8}\ \mathrm{m}$

gravity $t_{\mathrm{KH}} = \dfrac{GM_\odot^2}{R_\odot L_\odot} \approx 3 \times 10^{7}\ \mathrm{yr}$

Earth $4.5 \times 10^{9}\ \mathrm{yr}$ (radiometric)

fusion $4\,\mathrm{p} \to {}^4\mathrm{He}: \quad \Delta m/m = 0.71\%, \quad t_{\mathrm{nuc}} = \dfrac{0.1 \cdot 0.0071\, M_\odot c^2}{L_\odot} \approx 1 \times 10^{10}\ \mathrm{yr}$

$$t_{\mathrm{KH}} \ll 4.5\ \mathrm{Gyr} < t_{\mathrm{nuc}}$$

only nuclear energy lasts long enough

Eddington · Nature 106, 14 (1920) · Bethe · Phys. Rev. 55, 434 (1939)

3.2 Hydrostatic equilibrium and virial theorem

force balance $$\frac{\mathrm{d}p}{\mathrm{d}r} = -\frac{Gm(r)\rho}{r^2}$$

virial theorem $$2K + U = 0, \quad K = \frac{3}{2}Nk_B\overline{T}, \quad U \approx -\frac{3}{5}\frac{GM^2}{R}$$

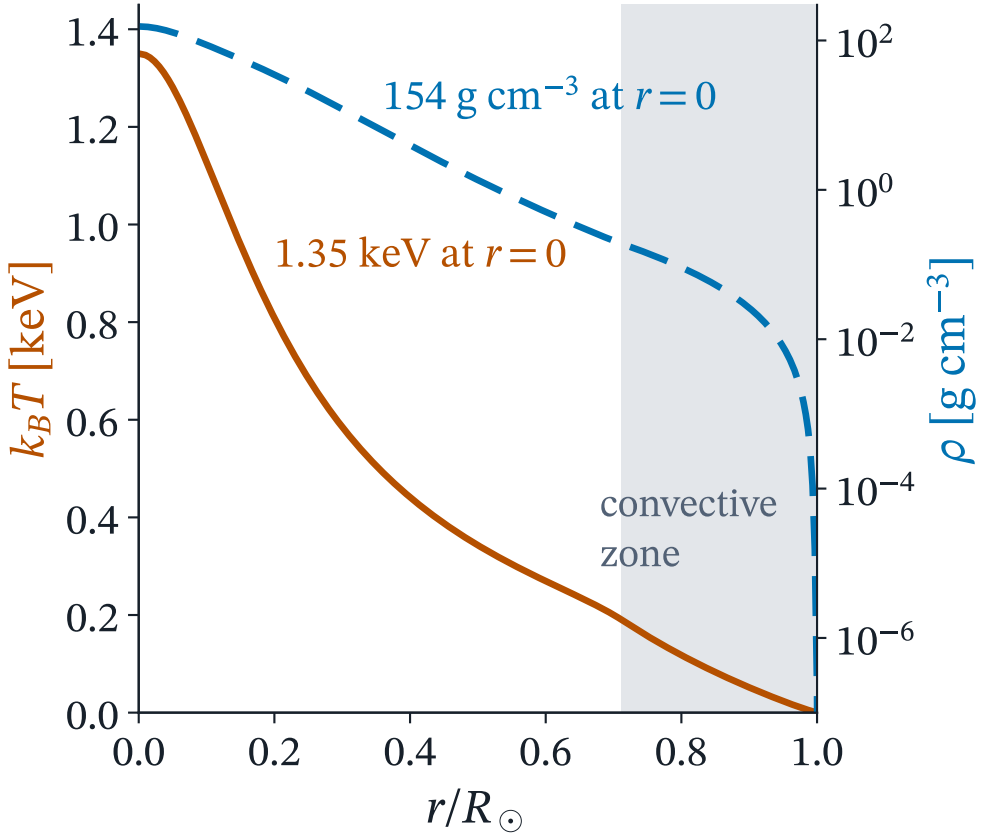
mean temperature $$k_B\overline{T} = \frac{GM\mu m_u}{5R} \approx 0.25\ \mathrm{keV} \quad (\mu = 0.62)$$

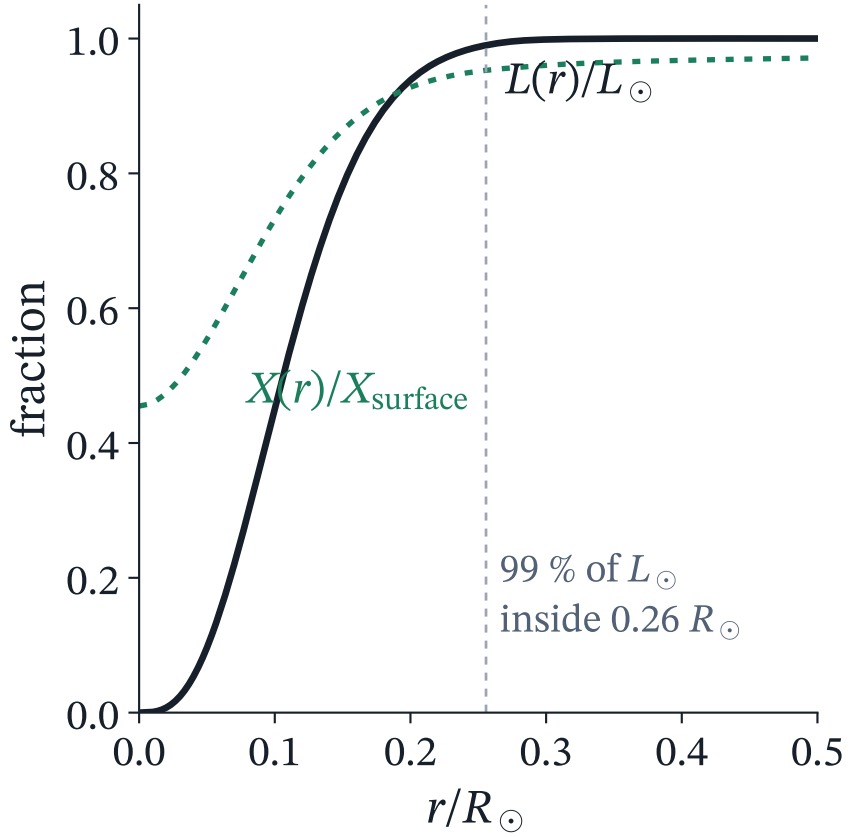
total energy $$E = K + U = -K \quad \Rightarrow \quad \frac{\mathrm{d}K}{\mathrm{d}E} = -1$$

losing energy heats the star

too much fusion: expands and cools · a thermostat

3.2 Inside the Sun





centre: $p_{\text{fus}} = 262\ \text{W/m}^3$ (a human body: about 1 kW/m^3)

Model S · Christensen-Dalsgaard et al. · Science 272, 1286 (1996)

3.2 The Sun and a tokamak

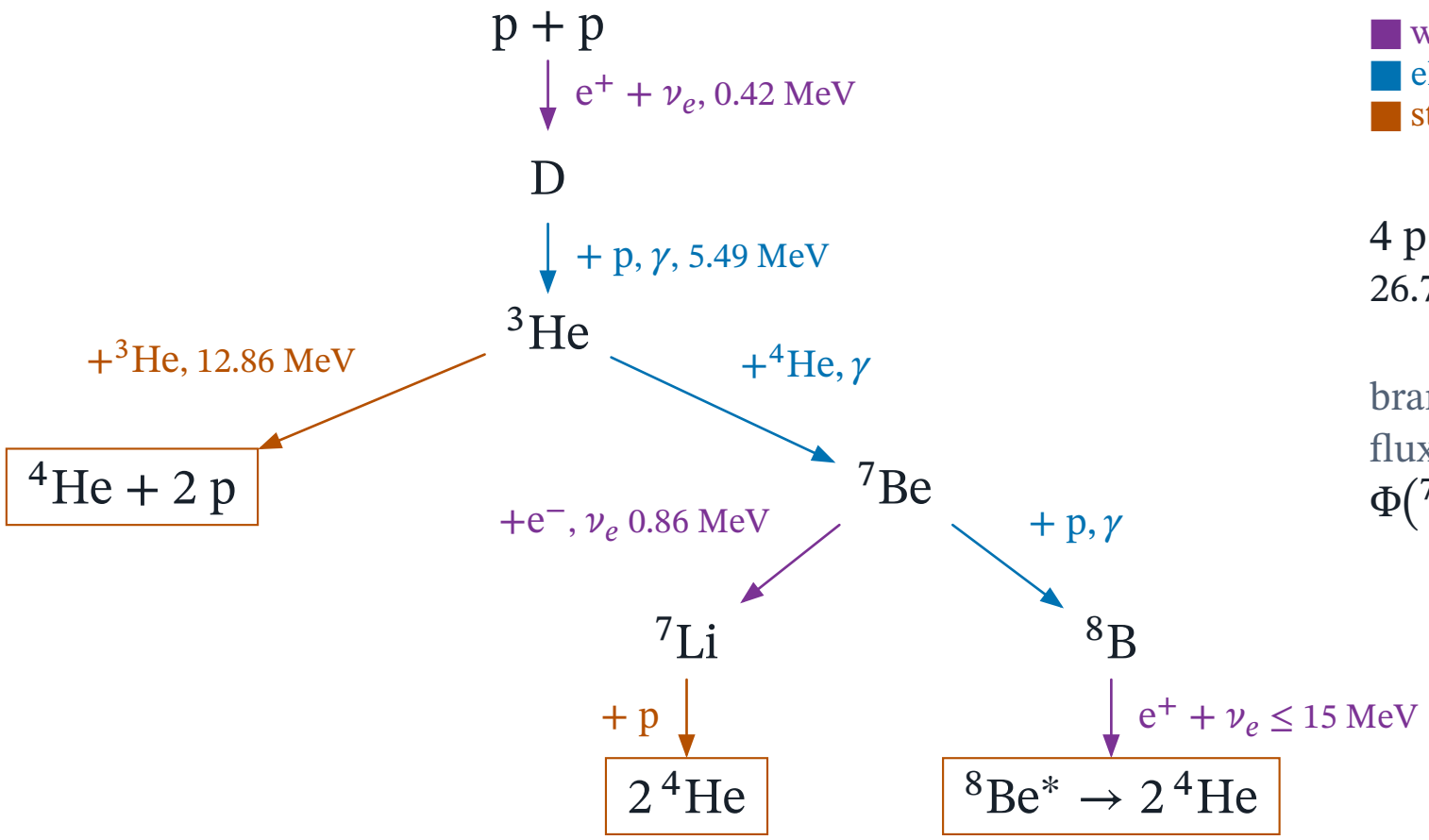
	Sun, centre	tokamak (ITER)
$k_B T$	1.35 keV	10–20 keV
n	$4.6 \times 10^{31}\,\mathrm{m}^{-3}$	$10^{20}\,\mathrm{m}^{-3}$
fuel	p–p (weak interaction)	D–T (strong)
p_{fus}	$262\ \mathrm{W/m^3}$	$0.6\ \mathrm{MW/m^3}$
$\tau_E = W_{\mathrm{th}}/L$	2×10^7 yr	3.7 s
$nT\tau_E$	$5 \times 10^{46}\,\mathrm{keV\,s\,m^{-3}}$	needs $3 \times 10^{21}\,\mathrm{keV\,s\,m^{-3}}$

the Sun wins by confinement time

not by reaction rate or power density

ITER Physics Basis · Nucl. Fusion 39, 2137 (1999)

3.3 Proton–proton chain



weak
electromagnetic
strong

$4\,\mathrm{p} \rightarrow {}^4\mathrm{He} + 2\mathrm{e}^+ + 2\nu_e$
26.7 MeV

branch shares from neutrino fluxes:
$\Phi({}^7\mathrm{Be})/\Phi(\mathrm{pp}) = f/(2-f)$

pp I · 84.8 %

pp II · 15.2 %

pp III · 0.017 %

Adelberger et al. · Rev. Mod. Phys. 83, 195 (2011) · Vinyoles et al. · ApJ 835, 202 (2017)

3.3 Which force sets the rate?

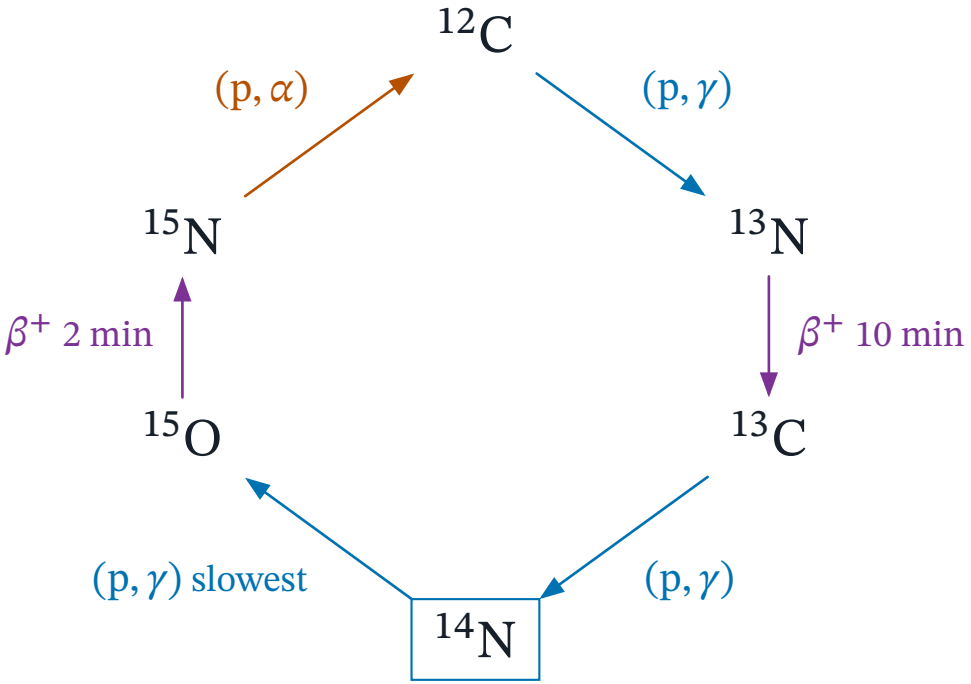
reaction	force	$S(0)$ [keV b]
$p + p \to D + e^+ + \nu$	weak	4.01×10^{-22}
$D + p \to {}^3He + \gamma$	electromagnetic	2.14×10^{-4}
${}^7Be + p \to {}^8B + \gamma$	electromagnetic	0.0208
${}^3He + {}^4He \to {}^7Be + \gamma$	electromagnetic	0.56
${}^{12}C + p \to {}^{13}N + \gamma$	electromagnetic	1.34
${}^{14}N + p \to {}^{15}O + \gamma$	electromagnetic	1.66
${}^3He + {}^3He \to {}^4He + 2p$	strong	5210

ν out: weak · γ out: electromagnetic · only nuclei: strong

the force sets S; the Coulomb barrier sets the temperature (Chapter 2)

Adelberger et al. · Rev. Mod. Phys. 83, 195 (2011), recommended $S(0)$

3.3 CNO cycle

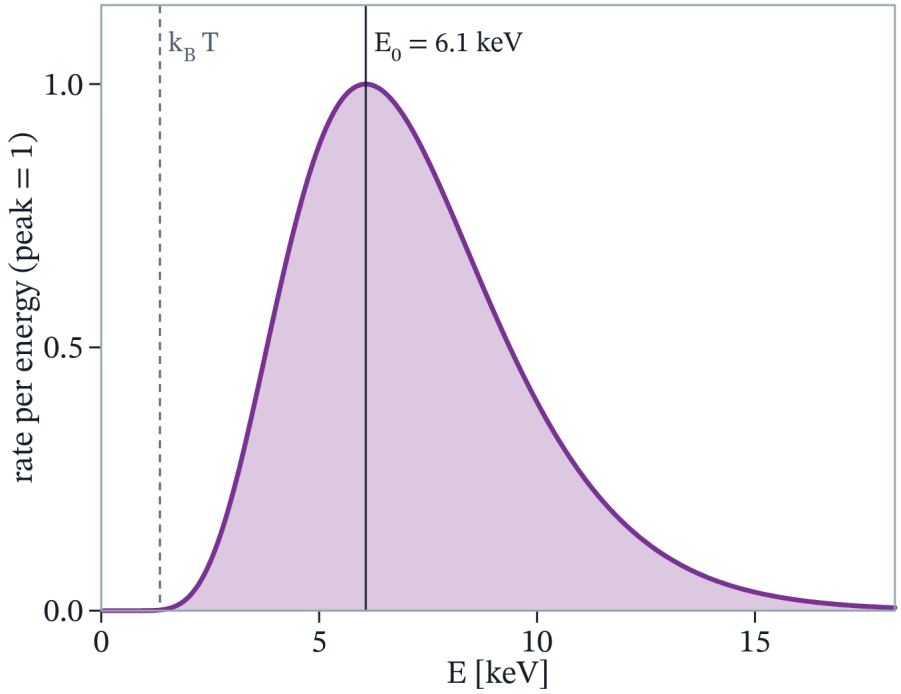


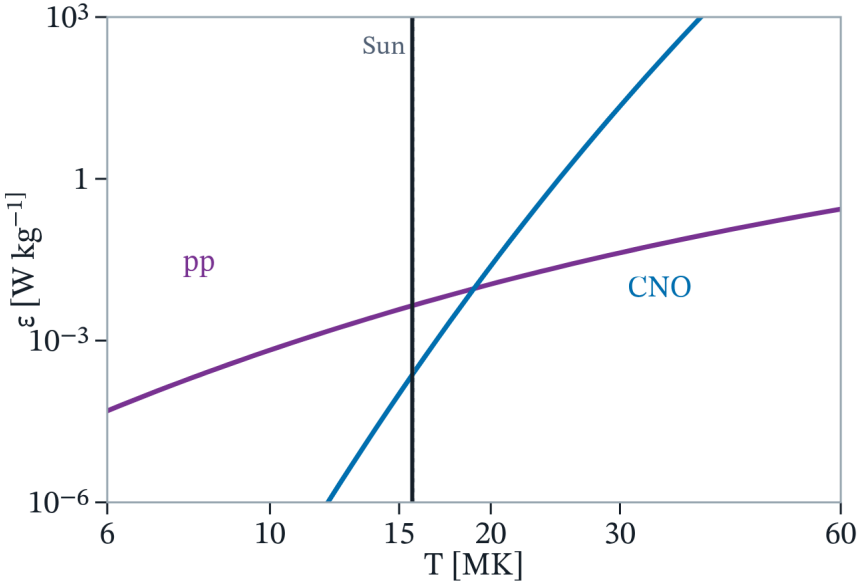
weak
electromagnetic
strong

bottleneck	$^{14}\mathrm{N} + \mathrm{p} : Z_1 Z_2 = 7$
net	$4\,\mathrm{p} \rightarrow {}^4\mathrm{He} + 2\mathrm{e}^+ + 2\nu_e$
energy	26.7 MeV, 25.0 MeV stays
catalyst	C, N, O are not used up

Bethe · Phys. Rev. 55, 434 (1939) · Adelberger et al. · Rev. Mod. Phys. 83, 195 (2011)

3.3 Gamow window of stellar fuels





pp: weak first step, S = 4 × 10⁻²² keV b · CNO: ¹⁴N(p,γ), electromagnetic, S = 1.66 keV b · Sun-centre density 1.5 × 10⁵ kg m⁻³, X = 0.7 · Adelberger et al., Rev. Mod. Phys. 83

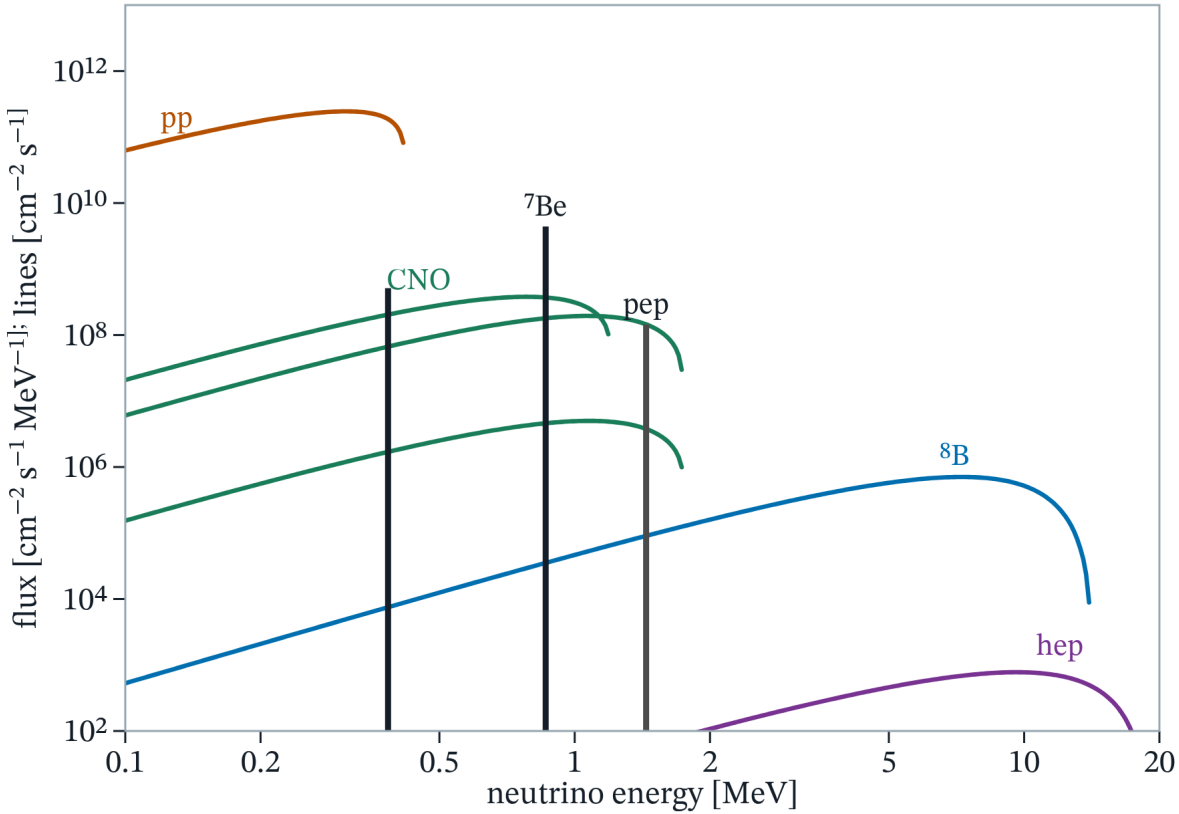
$E_G = 493$ keV, $E_0/k_BT = 4.5$, rate $\propto T^4$ · force: weak

burns H (pp chain) at 10–25 MK ✓

fuel ◉ p + p ○ p + ^{14}N ○ ^{4}He + ^{8}Be ○ ^{12}C + ^{12}C ○ ^{16}O + ^{16}O

T 15.6 MK = 1.35 keV

3.3 Solar neutrinos



all solar neutrinos

share of the flux (the count rate favours high energies)

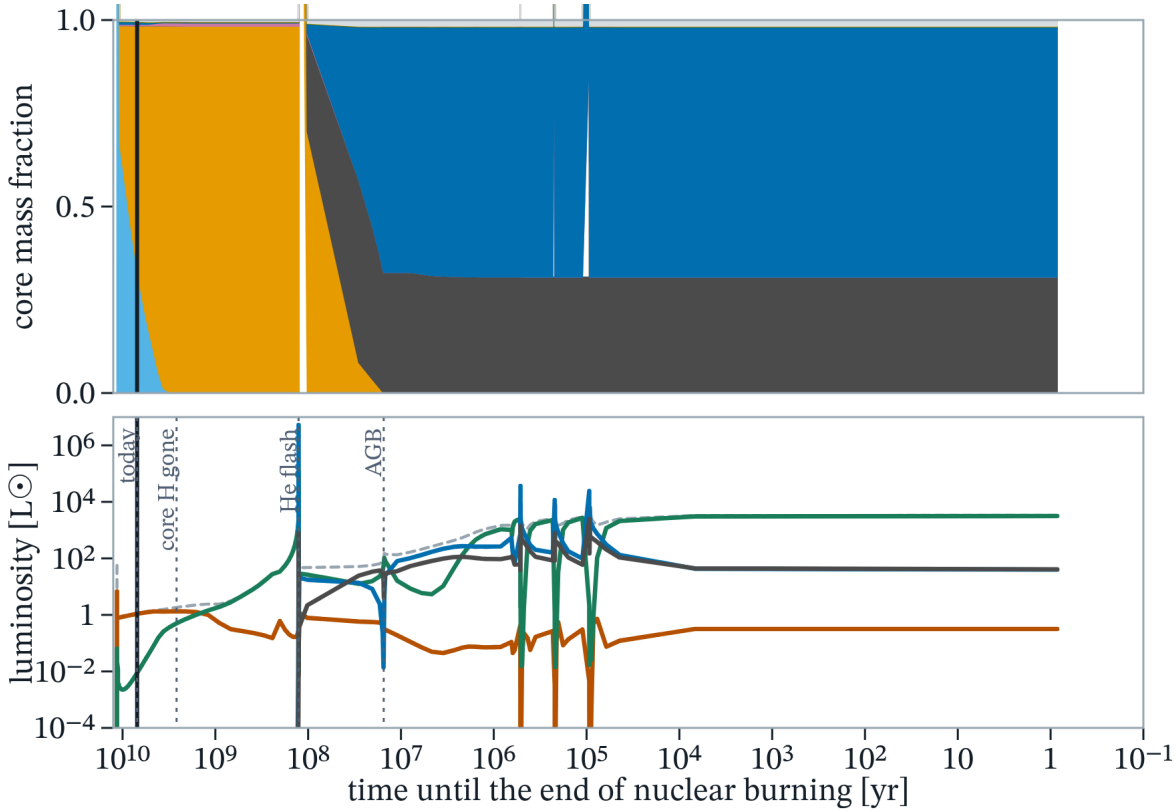
total 6.54×10^{10} cm^{-2} s^{-1}

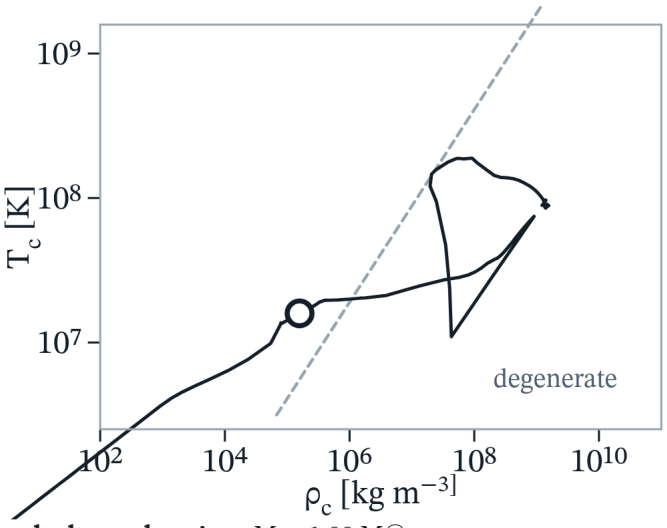
pp	91 %	pp chain
^{7}Be	8 %	pp II
^{13}N	0.43 %	CNO
^{15}O	0.31 %	CNO
pep	0.22 %	pp chain
^{8}B	0.01 %	pp III
^{17}F	0.01 %	CNO
hep	0.00 %	pp chain

all from weak reactions · B16-GS98 fluxes, Vinyoles et al. 2017 · Ga: GALLEX/SAGE, Cl: Homestake, water: Super-Kamiokande/SNO

detector ◉ none ○ Ga (GALLEX, SAGE) ○ Cl (Homestake) ○ water Cherenkov (Super-K, SNO)

3.3 The life of the Sun





core hydrogen burning · M = 1.00 M⊙

T_c = 15.9 MK = 1.37 keV, $\rho_c = 1.6 \times 10^5$ kg m^{-3}

CNO share of hydrogen burning: 0.8 %

end: carbon–oxygen white dwarf of 0.54 M⊙

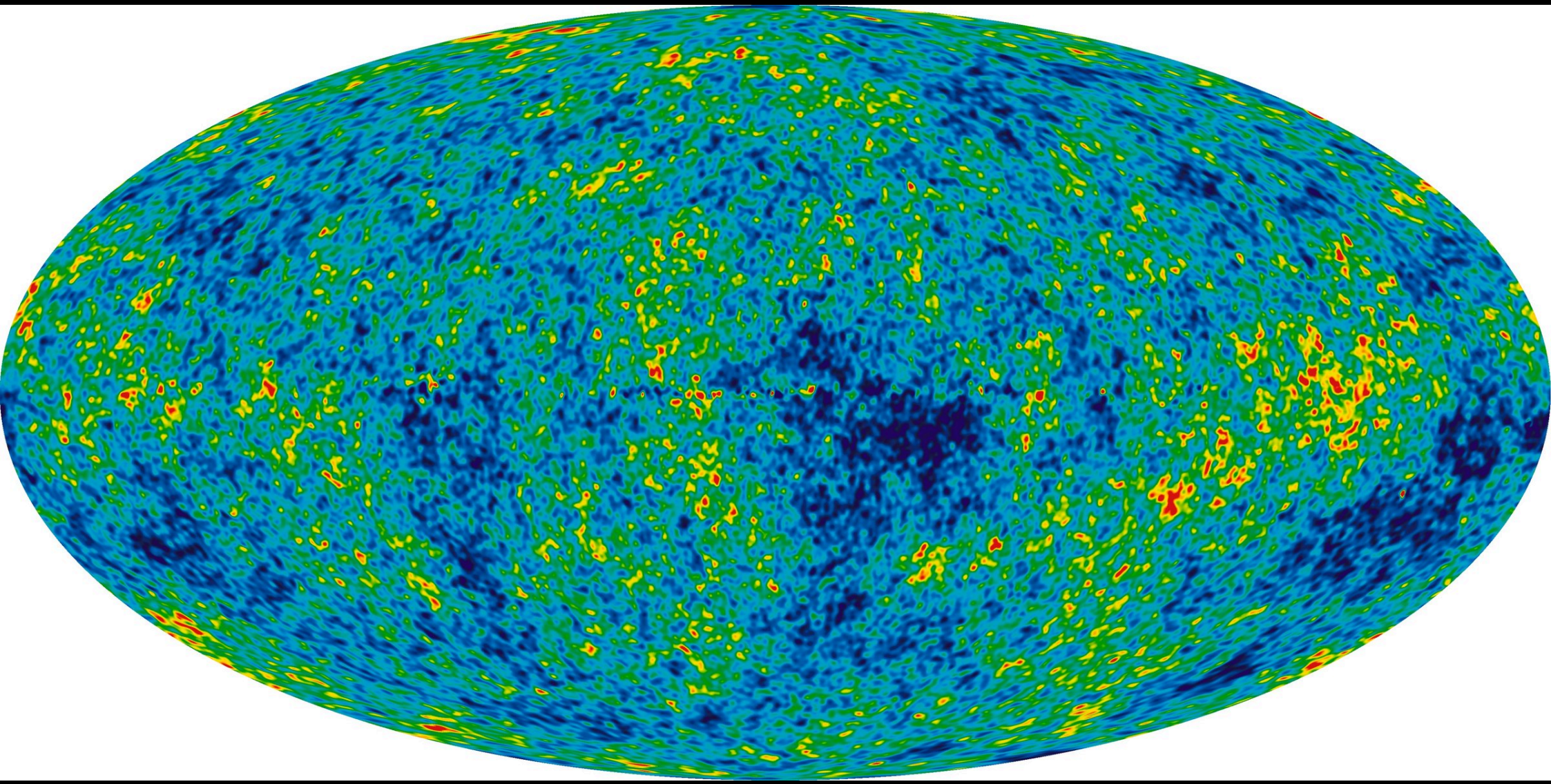
H He C N O Ne Mg Si

pp chain CNO cycle triple alpha carbon burning

MIST v1.2 (MESA) tracks · Dotter 2016, Choi et al. 2016

age 4.50 Gyr

run ☐



NASA / WMAP Science Team, public domain

3.4 Neutrons decide the helium

freeze-out $\quad \mathrm{n} + \nu_e \rightleftarrows \mathrm{p} + \mathrm{e}^-$ (weak): $n/p = e^{-Q/k_B T_f}, \quad Q = 1.293\ \mathrm{MeV}, \quad k_B T_f \approx 0.8\ \mathrm{MeV} \Rightarrow 0.20$

bottleneck $\quad \mathrm{n} + \mathrm{p} \to \mathrm{D} + \gamma$ (e.m.): $n_\mathrm{D}/n_\mathrm{n} \approx \frac{3}{4}\eta \frac{2\zeta(3)}{\pi^2}\left(\frac{4\pi k_B T}{mc^2}\right)^{3/2} e^{B_\mathrm{D}/k_B T} = 1 \Rightarrow k_B T_\mathrm{D} = 0.066\ \mathrm{MeV}$

decay $\quad n/p = 0.20\, e^{-\Delta t/\tau_n} = 0.14 \quad (\tau_n = 878\ \mathrm{s}, t_\mathrm{D} \approx 300\ \mathrm{s})$

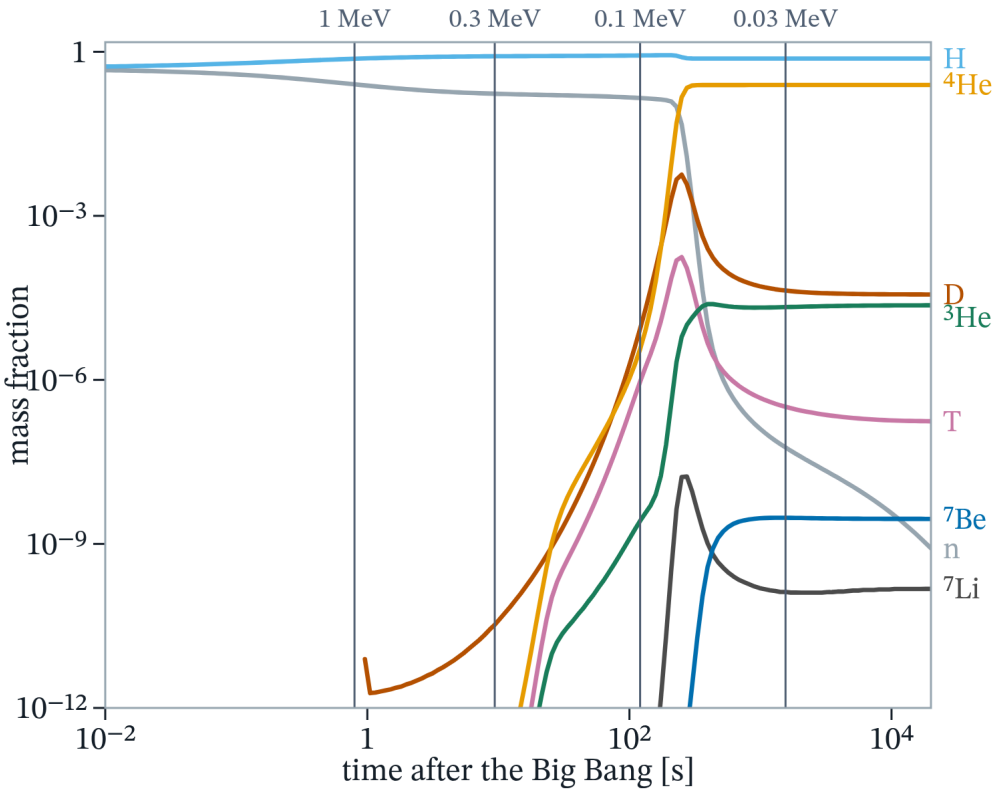
helium $\quad Y_p = \frac{2n/p}{1 + n/p} = 0.25$

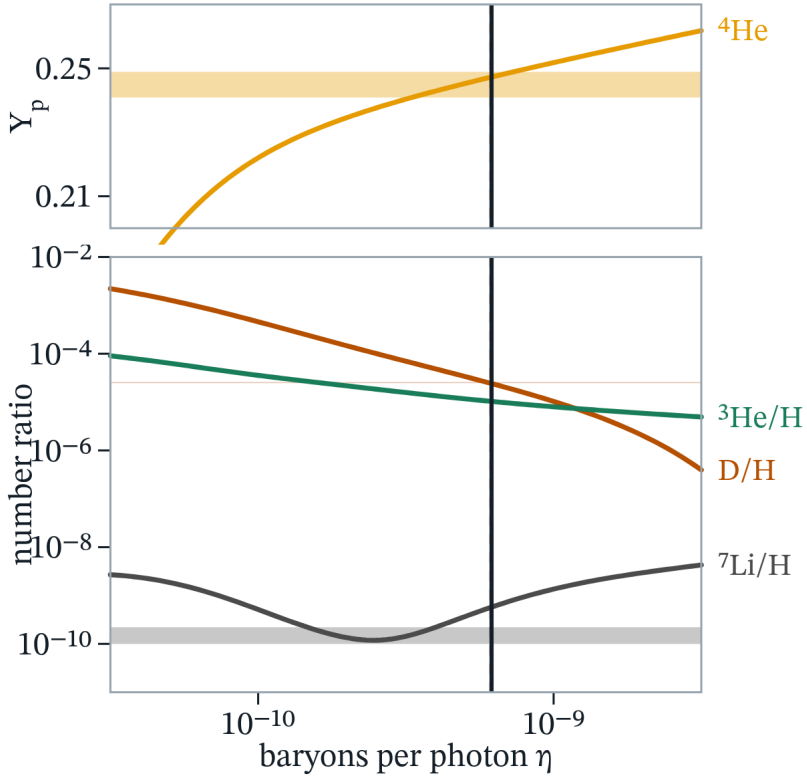
$$Y_p = 0.247$$

network (LINX) · observed 0.245 ± 0.004

Alpher, Bethe & Gamow · Phys. Rev. 73, 803 (1948) · LINX · Phys. Rev. D 112, 063531 (2025) · Aver et al. · JCAP 07, 011 (2015)

3.4 Primordial nucleosynthesis





$Y_p = 0.2471 \cdot D/H = 2.48 \times 10^{-5} \cdot {}^7Li/H = 5.57 \times 10^{-10}$

3 neutrino species ($N_{eff} = 3.045$)

bands: observed (±2σ for D, Li) · dashed: η from the CMB (Planck) · LINX, PRIMAT 2023 network

η $\eta = 6.17 \times 10^{-10}$

neutrino species ○ 2 ◉ 3 ○ 4 ○ 5

3.4 Why it stops at lithium

no stable $A = 5, 8$ $^5\mathrm{He}, ^5\mathrm{Li} : 10^{-21}$ s, $^8\mathrm{Be} : 8 \times 10^{-17}$ s

dilute $\rho_b \approx 10^{-2}$ kg/m^3 at $k_B T = 0.07$ MeV (Sun: 1.5×10^5)

fast expansion: $T \propto t^{-1/2}$, over after about 20 min

$^7\mathrm{Li}/\mathrm{H}$: predicted 5×10^{-10}, observed 1.6×10^{-10}

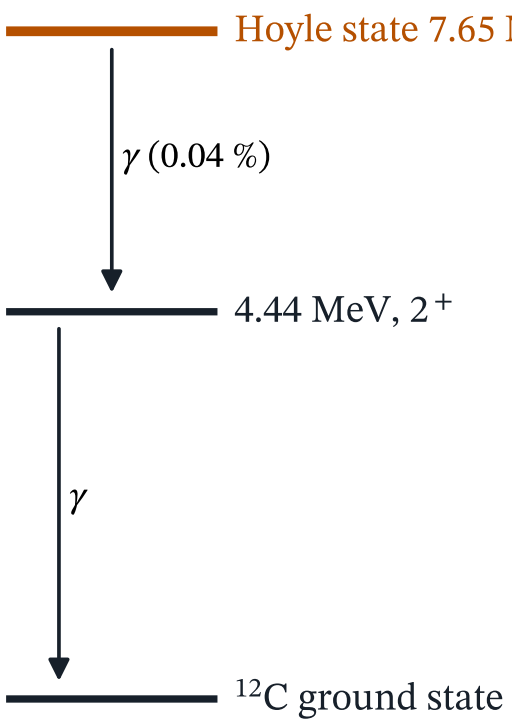
the lithium problem

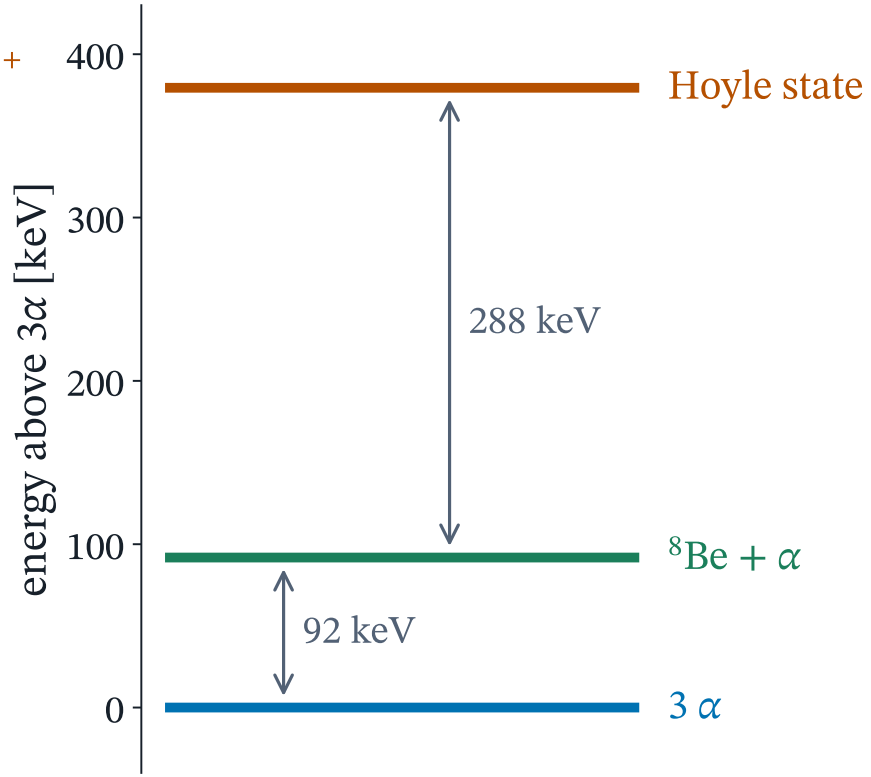
Cooke et al. · ApJ 855, 102 (2018) · PDG BBN review (2024)



NASA, ESA, N. Smith (UC Berkeley), Hubble Heritage Team (STScI/AURA), CC BY 4.0

3.5 Triple alpha and the Hoyle state





$^4\mathrm{He} + {}^4\mathrm{He} \rightleftarrows {}^8\mathrm{Be}$ strong

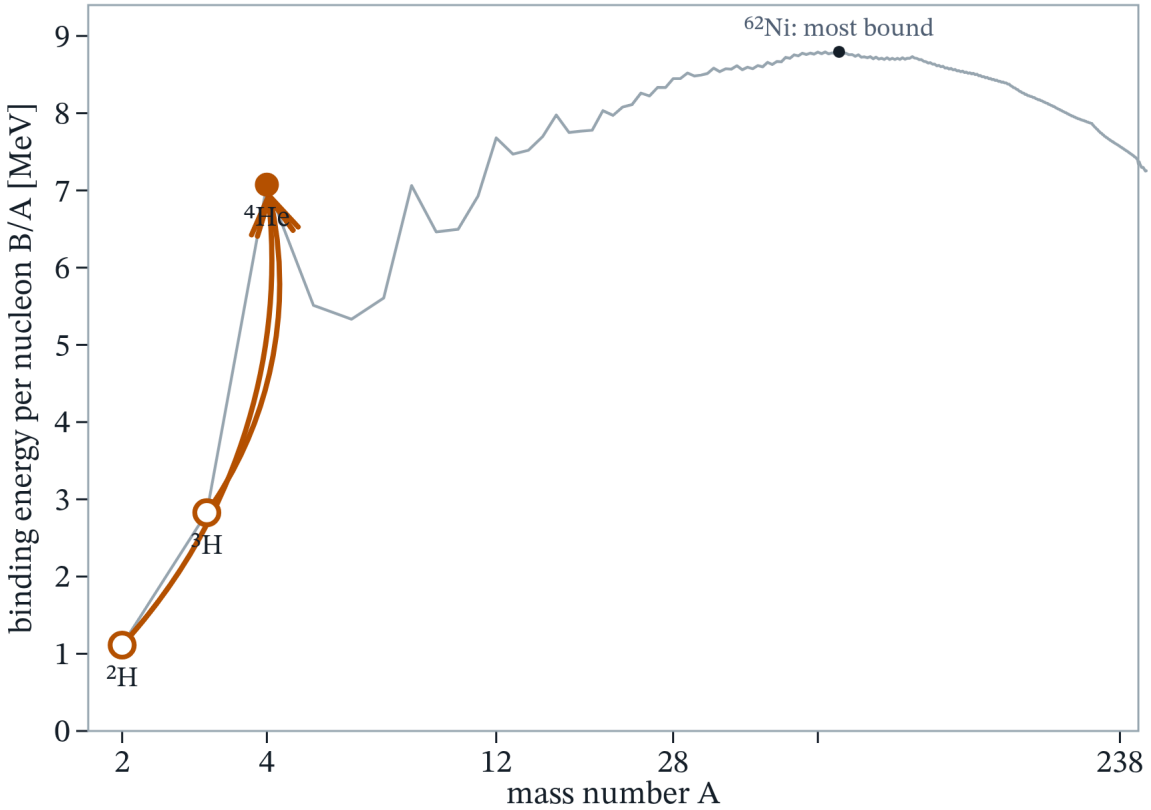
$^8\mathrm{Be} + {}^4\mathrm{He} \rightarrow {}^{12}\mathrm{C}^*$ resonance

$^{12}\mathrm{C}^* \rightarrow {}^{12}\mathrm{C} + \gamma$ e.m.

$^{12}\mathrm{C} + {}^4\mathrm{He} \rightarrow {}^{16}\mathrm{O} + \gamma$ e.m.

Hoyle · ApJS 1, 121 (1954) · Cook, Fowler, Lauritsen & Lauritsen · Phys. Rev. 107, 508 (1957) · AME2020

3.5 Energy per nucleon



D + T → ^{4}He + n

Q = 17.59 MeV · 3.518 MeV per nucleon

33.9 × 10^{13} J per kg of fuel

= 8.1 × 10^{4} t of TNT per kg; coal: 3 × 10^{7} J/kg

force: strong

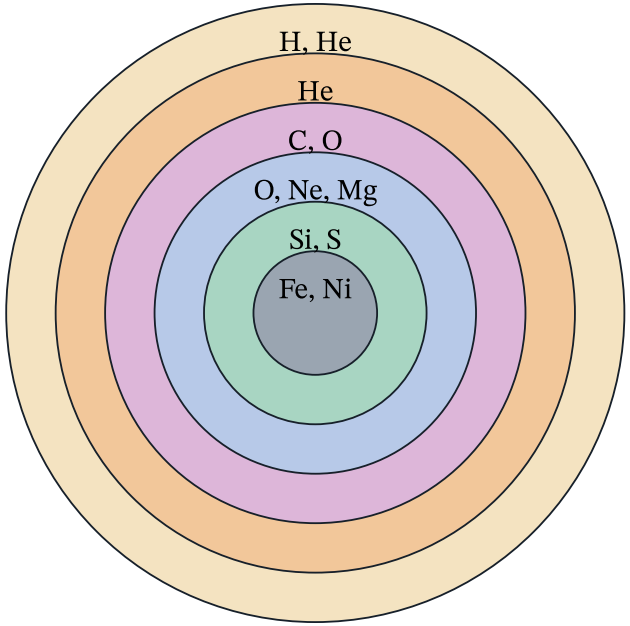
on Earth: 10–20 keV

open: fuel · filled: ash · Q from AME2020 atomic masses (Wang et al. 2021)

- (●) D + T → ^{4}He + n
- ( ) 4 p → ^{4}He
- ( ) 3 ^{4}He → ^{12}C
- ( ) ^{12}C + ^{4}He → ^{16}O
- ( ) 2 ^{12}C → ^{20}Ne + ^{4}He
- ( ) 2 ^{16}O → ^{28}Si + ^{4}He
- ( ) 2 ^{28}Si → ^{56}Ni (net)
- ( ) n + ^{235}U → ^{141}Ba + ^{92}Kr + 3 n

3.5 Burning stages of a massive star

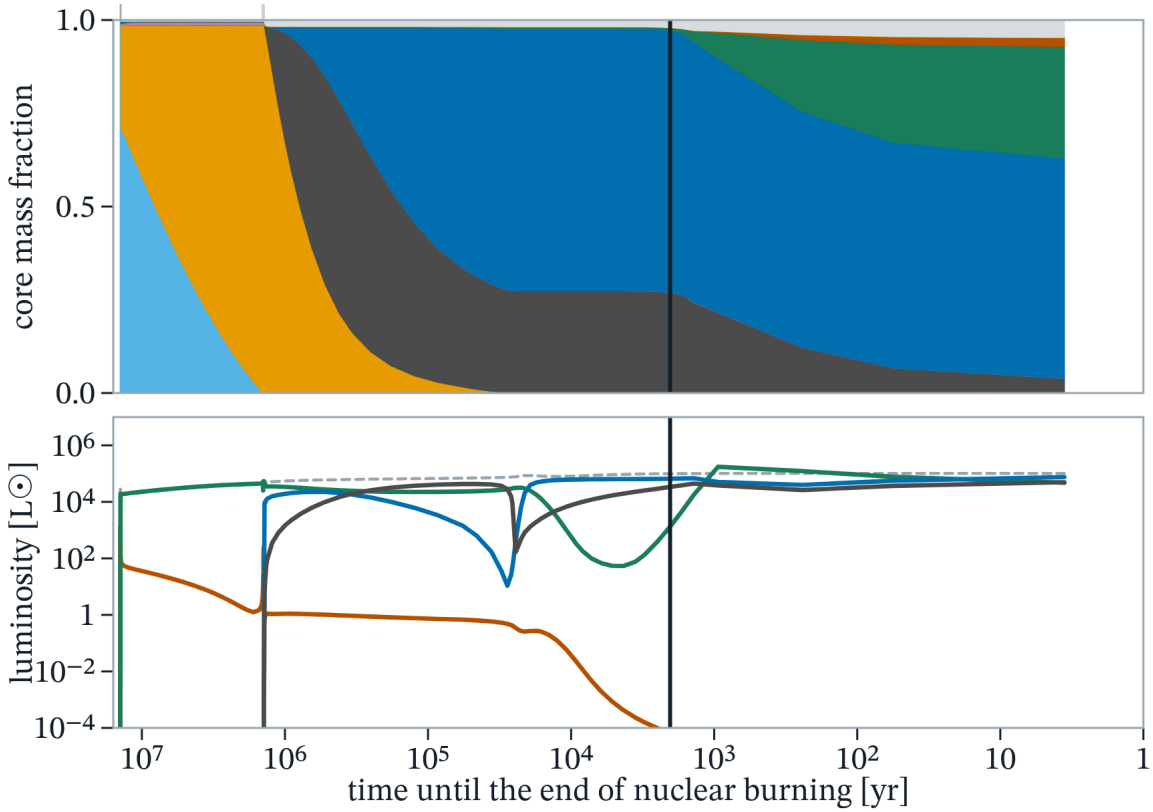
fuel	ash	T [GK]	$k_B T$ [keV]	duration
H	He	0.037	3	8.1 Myr
He	C, O	0.19	16	1.2 Myr
C	Ne, Mg	0.87	75	980 yr
Ne	O, Mg	1.6	140	0.6 yr
O	Si, S	2.0	170	1.3 yr
Si	Fe, Ni	3.3	280	11 d

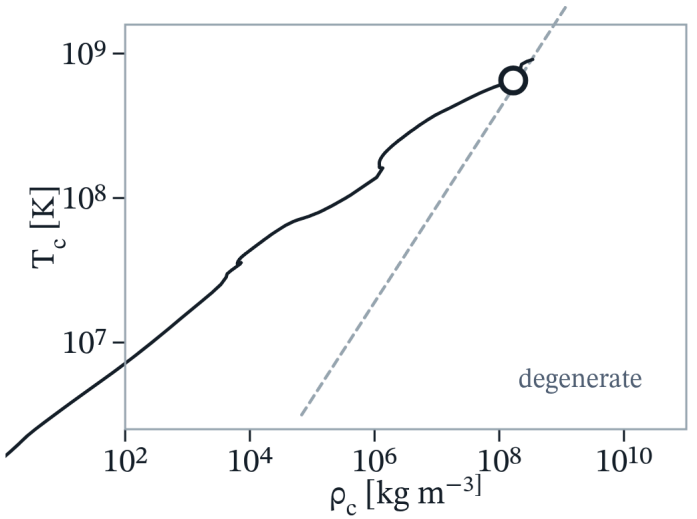


20 $M_\odot$ after helium: neutrino losses $\propto T^9$

Heger, Woosley, Fryer & Langer · Massive star evolution through the ages (2003), Table 1 · shells not to scale

3.5 Build a star





shell burning around an inert core · M = 11.56 M⊙

T_c = 652 MK = 56.15 keV, $\rho_c = 1.7 \times 10^8$ kg m^{-3}

end: C exhausted; Ne, O, Si burn within about 2 years, the Fe core collapses: supernova, neutron star or black hole

then (20 M⊙ model): Ne 0.6 yr, O 1.3 yr, Si 11 d

H He C N O Ne Mg Si

pp chain CNO cycle triple alpha carbon burning

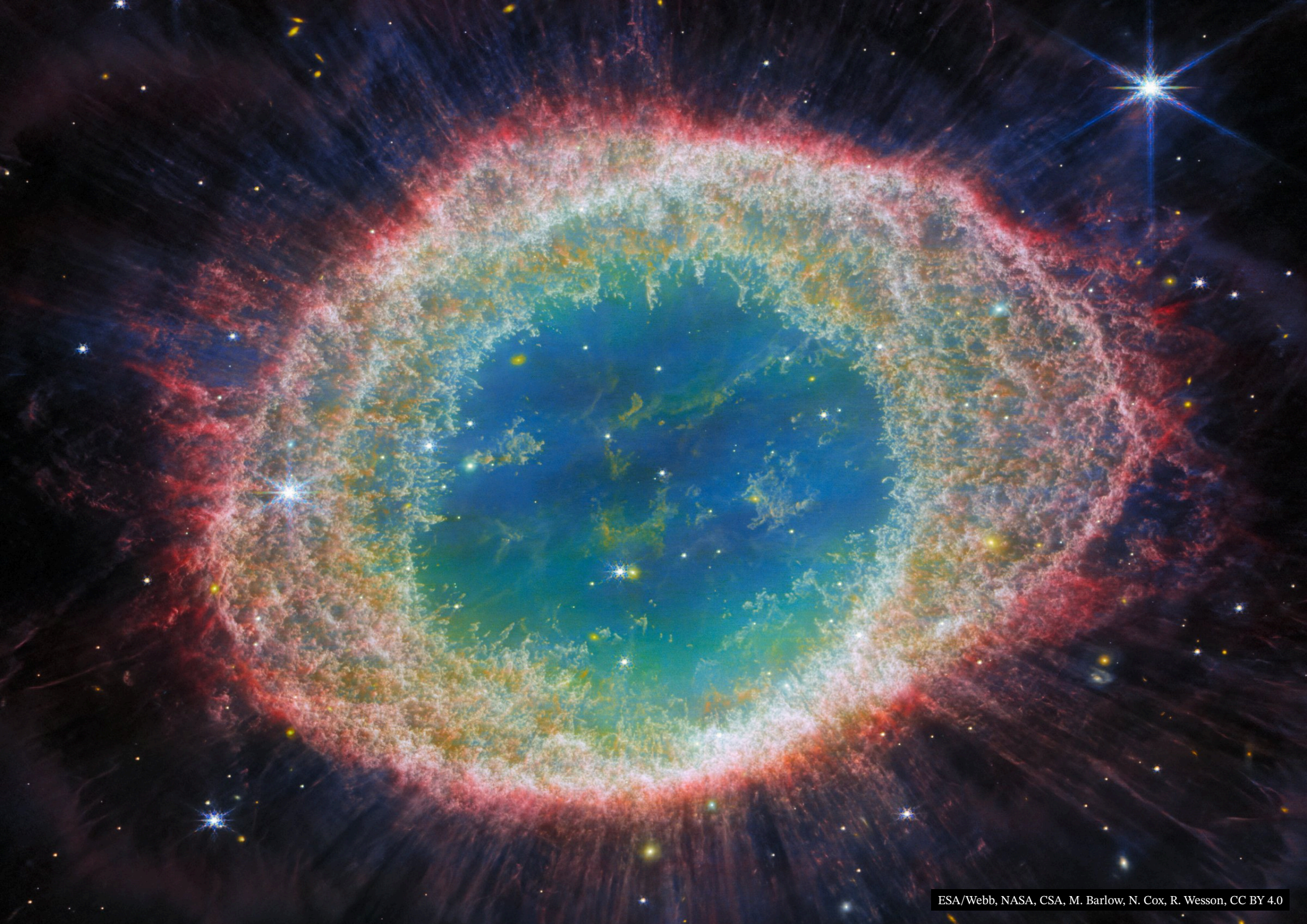
MIST v1.2 (MESA) tracks · Dotter 2016, Choi et al. 2016

initial mass — 15 M⊙

metals — ○ Z = 0.00014 ○ Z = 0.001 ◉ Z = 0.014 (Sun) ○ Z = 0.045

age — 14.17 Myr

run — ☐



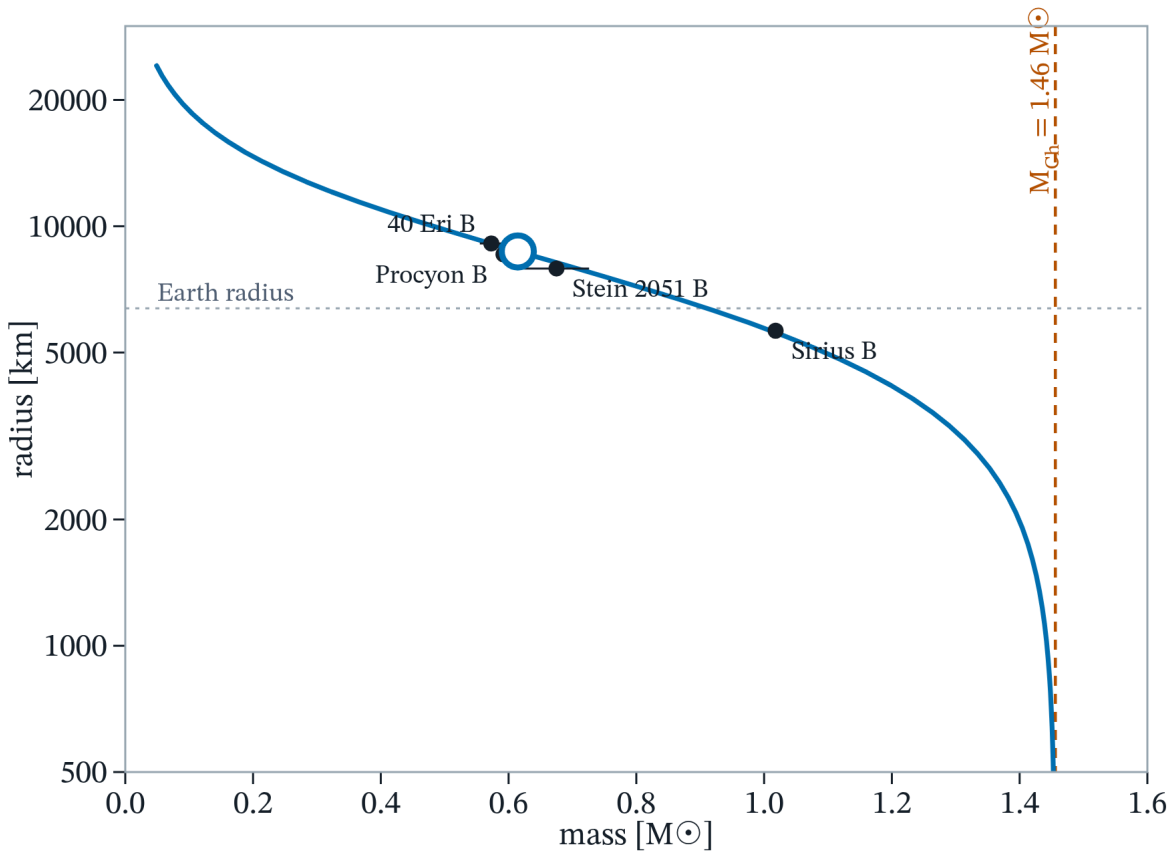
ESA/Webb, NASA, CSA, M. Barlow, N. Cox, R. Wesson, CC BY 4.0

3.6 Degeneracy ends the burning

degenerate	$E_F > k_B T$: pressure no longer depends on T, no thermostat
lightest star	$0.075 M_\odot$: degeneracy stops the contraction before p–p ignites
brown dwarf	$0.013 M_\odot \approx 13 M_{\text{Jupiter}}$: deuterium burning only
Chandrasekhar	$M_{\text{Ch}} = \frac{\sqrt{3\pi}}{2}\,\omega_3 \left(\frac{\hbar c}{G}\right)^{3/2} \frac{1}{(\mu_e m_u)^2} = 1.46 M_\odot \quad (\mu_e = 2)$
neutron star	$1.4 M_\odot$, $R \approx 12$ km, nuclear density

Chandrasekhar · ApJ 74, 81 (1931) · Chabrier & Baraffe · ARA&A 38, 337 (2000) · Spiegel et al. · ApJ 727, 57 (2011)

3.6 White dwarfs



$M = 0.614\ \mathrm{M}\odot$, $R = 8708$ km

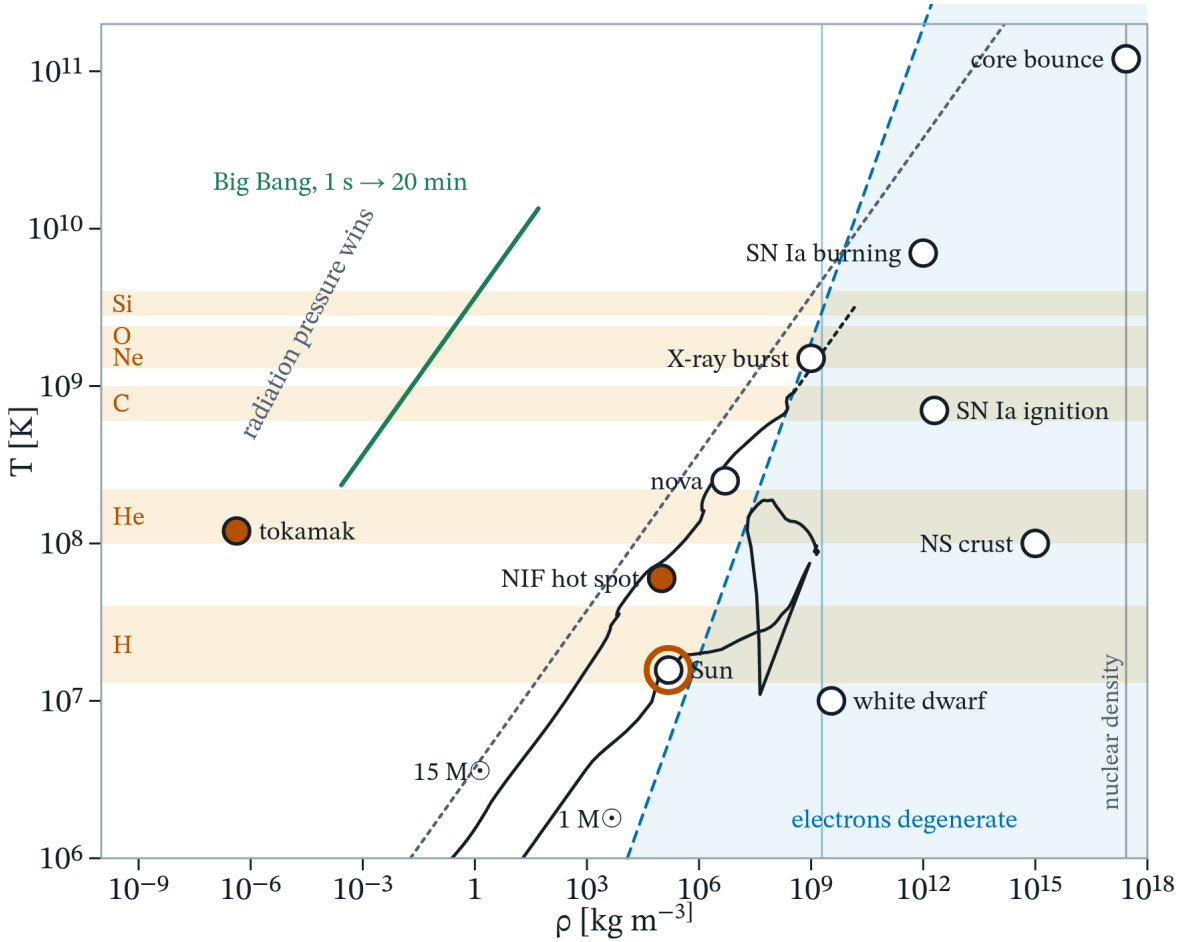
$p_\mathrm{F}/m_\mathrm{e}c = 1.2$ · electrons becoming relativistic

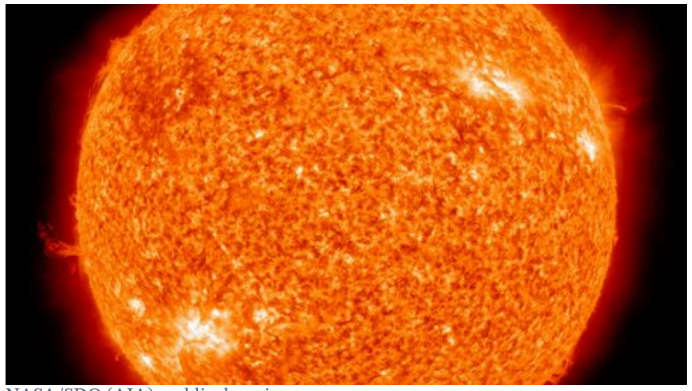
more mass, smaller star; at M_Ch the radius goes to zero

curve: cold electron gas, $\mu_\mathrm{e} = 2$ (no Coulomb or thermal corrections) · points: dynamical masses and radii, Bond et al., ApJ 848, 16 (2017), Table 2

ρ_c $3.5 \times 10^9\ \mathrm{kg\ m^{-3}}$

3.6 Where fusion happens





NASA/SDO (AIA), public domain

Sun, centre

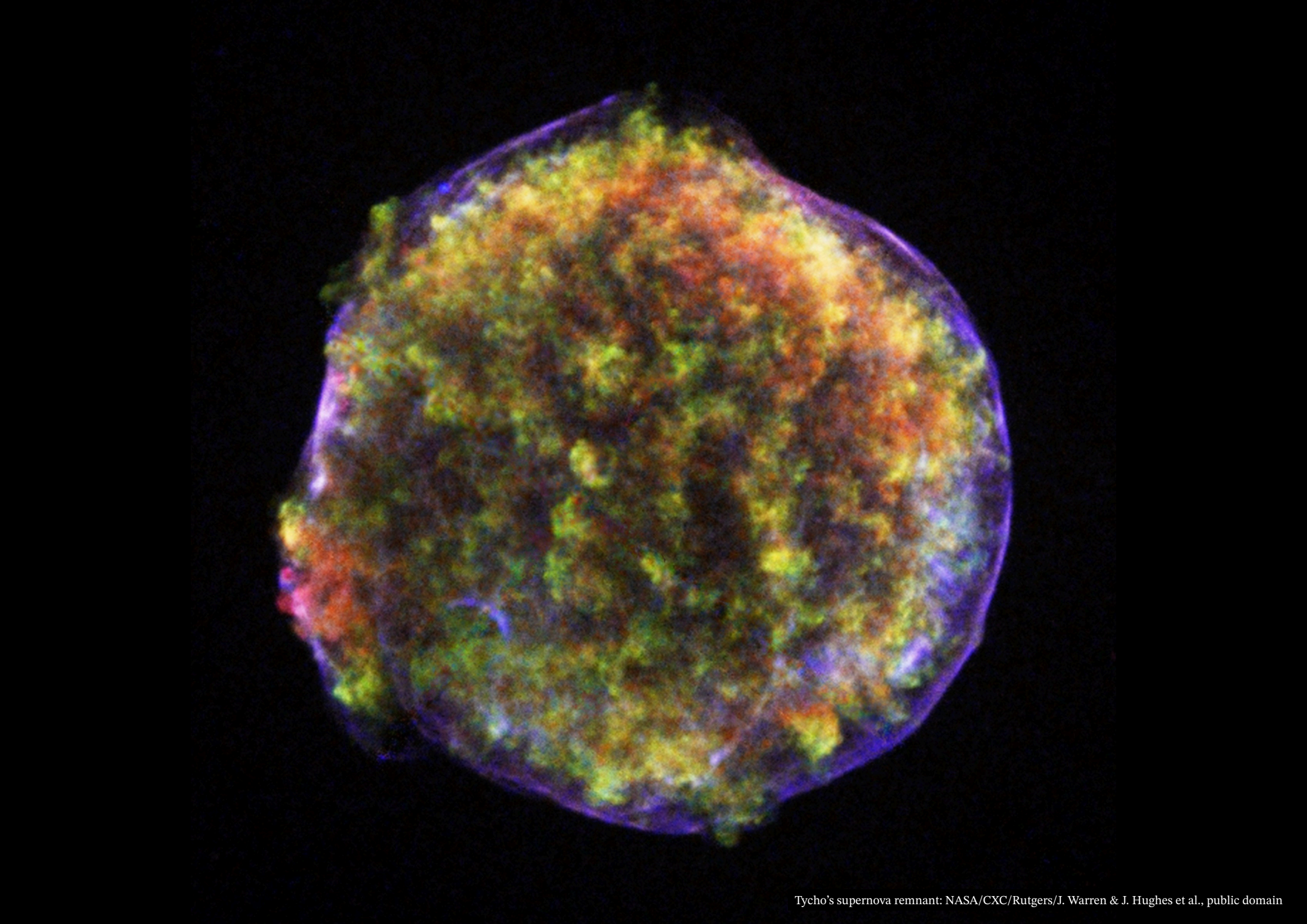
$\rho \approx 1.5 \times 10^{5}$ kg m^{-3}, $T \approx 1.6 \times 10^{7}$ K

pp chain (99 %), CNO (1 %)

Model S, Christensen-Dalsgaard et al., Science 272, 1286 (1996), doi:10.1126/science.272.5266.1286

orange bands: burning temperatures (20 M☉, Heger et al. 2003) · solid: MIST centres · dotted: $\rho \propto T^3$

◉ Sun ○ white dwarf ○ nova ○ X-ray burst ○ SN Ia ignition ○ SN Ia burning ○ core bounce ○ NS crust ○ tokamak ○ NIF hot spot



Tycho’s supernova remnant: NASA/CXC/Rutgers/J. Warren & J. Hughes et al., public domain

3.6 Type Ia supernova: a carbon bomb

progenitor — C/O white dwarf grows towards M_{Ch}

ignition — $^{12}\mathrm{C} + {}^{12}\mathrm{C}$ (strong) at $\rho_c \approx 2 \times 10^{12}$ kg/m^3: degenerate, runs away

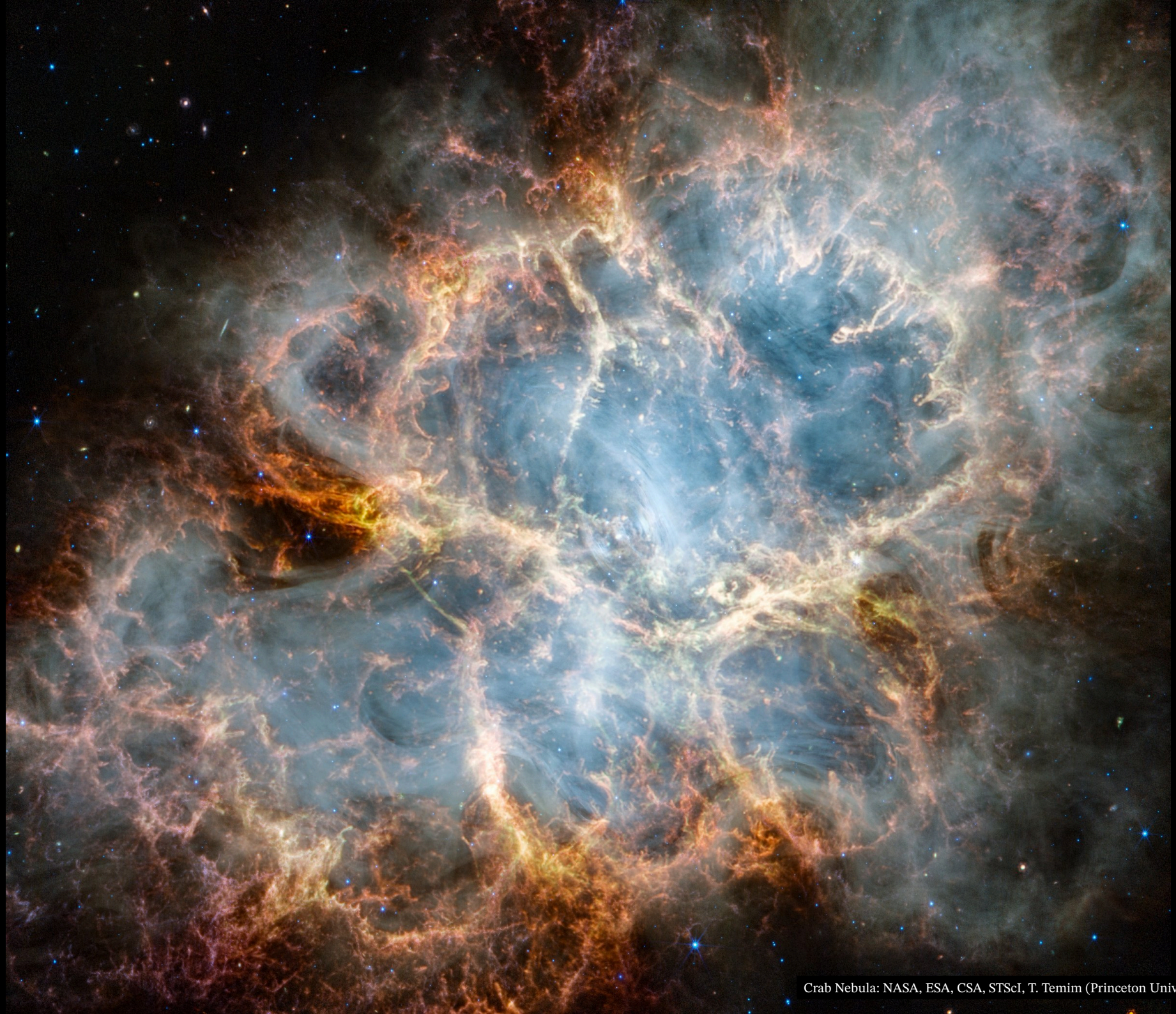
energy — C, O $\rightarrow {}^{56}\mathrm{Ni}$: 0.81 MeV per nucleon $\times\, 1.4 M_\odot = 2 \times 10^{44}$ J > binding

light — $^{56}\mathrm{Ni} \xrightarrow{6.1\,\mathrm{d}} {}^{56}\mathrm{Co} \xrightarrow{77\,\mathrm{d}} {}^{56}\mathrm{Fe}$ (weak)

fusion unbinds the whole star

no remnant · about 0.6 $M_\odot$ of $^{56}\mathrm{Ni}$

Hillebrandt & Niemeyer · ARA&A 38, 191 (2000)



Crab Nebula: NASA, ESA, CSA, STScI, T. Temim (Princeton University), CC BY 4.0

3.6 Core collapse: gravity, not fusion

iron core	no fusion energy left: B/A is maximal
collapse	photodisintegration, electron capture (weak) bounce at nuclear density
energy	$\frac{3}{5}\frac{GM^2}{R} = 3 \times 10^{46}$ J (1.4 $M_\odot$, 12 km), 99 % neutrinos
SN 1987A	24 neutrinos in Kamiokande-II, IMB, Baksan; $0.07 M_\odot$ of ^{56}Ni

fusion builds the core and burns the ejecta

explosive Si and O burning

Hirata et al. · PRL 58, 1490 (1987) · Bionta et al. · PRL 58, 1494 (1987) · Arnett et al. · ARA&A 27, 629 (1989) · Janka et al. · Phys. Rep. 442, 38 (2007)

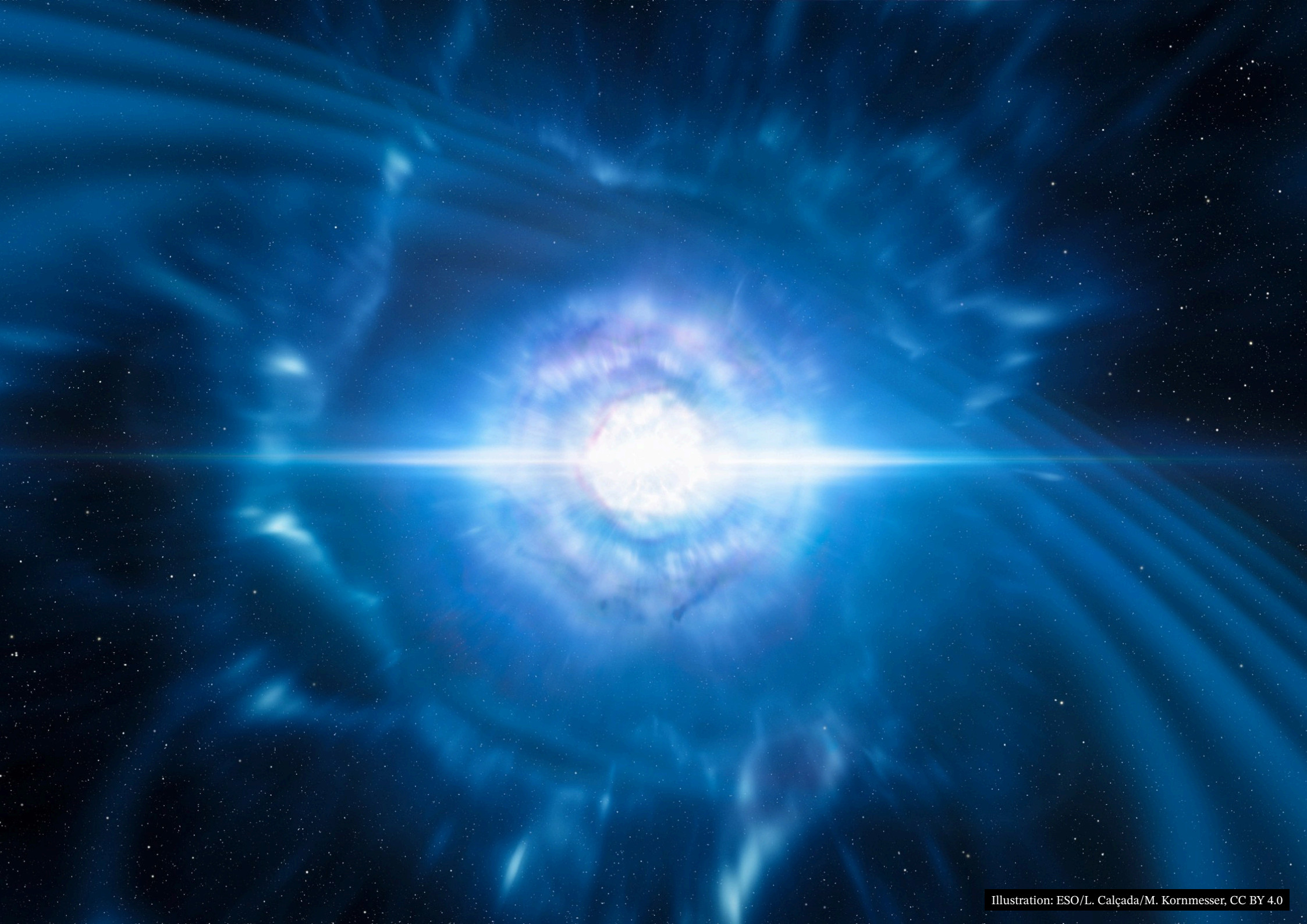


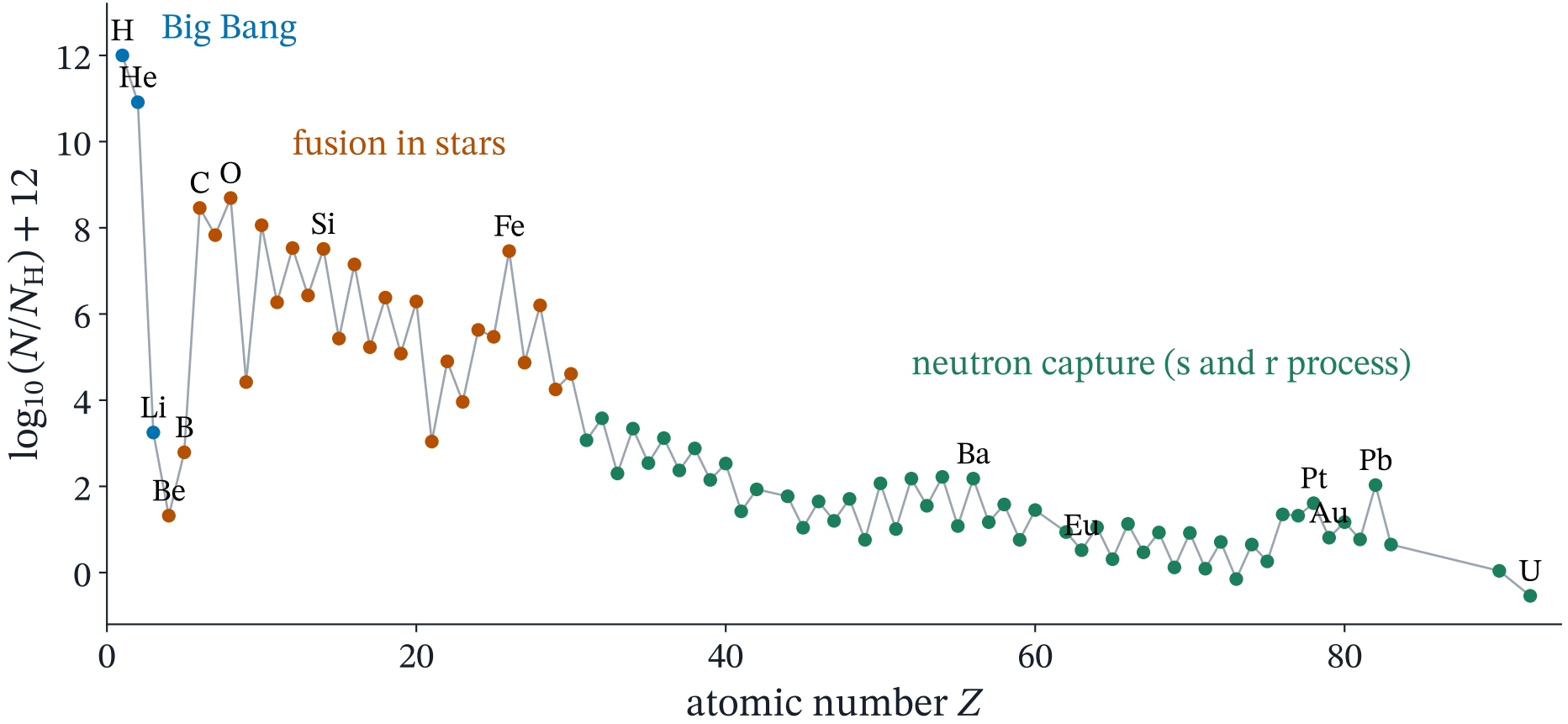
Illustration: ESO/L. Calçada/M. Kornmesser, CC BY 4.0

3.6 Neutron stars: bursts, crusts, mergers

X-ray burst	accreted H, He ignite: He flash, rapid proton capture (e.m.) up to Sn–Te in seconds
crust	pycnonuclear fusion: zero-point motion in a lattice, density instead of temperature
merger	GW170817: neutron capture (r process), kilonova · not fusion
on Earth	muon-catalysed fusion is fusion by density too (Chapter 9)

Schatz et al. · PRL 86, 3471 (2001) · Haensel & Zdunik · A&A 227, 431 (1990) · LIGO/Virgo · PRL 119, 161101 (2017)

3.6 Where the elements come from



Asplund, Amarsi & Grevesse · A&A 653, A141 (2021), CC BY 4.0 · Burbidge, Burbidge, Fowler & Hoyle · Rev. Mod. Phys. 29, 547 (1957)

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Data: MIST (Dotter 2016; Choi et al. 2016), LINX/PRIMAT, Model S, AME2020, Asplund et al. 2021, B16, Adelberger et al. 2011, Bond et al. 2017, Heger et al. 2003.
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AI tools assisted drafting and layout; scientific review by Christopher Albert.